Five Common IV Designs

BUSS975: Causal Inference in Financial Research

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Where we left off

- We built the machinery: the Wald ratio, 2SLS, weak instruments, and the five LATE assumptions. This one is about **where IVs actually come from**.
- IV is a general-purpose strategy: it works wherever a credible shock turns up. But over the years certain strategies have been used so often that they have become **designs** — each with its own literature and its own conventions.
- Knowing the designs tells you where to look for an IV in your own work.

The five

1. **Lotteries** — randomised offers with imperfect take-up.
2. **Judge / examiner fixed effects** (the "leniency design").
3. **Bartik, or shift-share, IVs.**
4. **Fuzzy regression discontinuity** — which you already know.
5. **Supply shifters for the elasticity of demand** — the original IV, from the 1920s.

- For each one, ask the same four questions:

What is the shock? *What is the first stage?*

Who are the compliers? *Where does exclusion break?*

- The last question is the one that decides whether the paper survives refereeing.

1. Lotteries

The design: randomisation is not enough

- In many randomised trials, assignment is random but **take-up is voluntary**.
- Those who take it up are precisely those who expect to benefit — **positive selection**.
- So comparing the treated with the untreated is biased **even inside a randomised experiment**.

- **The fix:** instrument actual treatment with the **randomised offer**, and estimate the LATE.

Oregon Health Insurance Experiment

- In 2008 Oregon expanded Medicaid for low-income adults. Facing excess demand, it allocated places by **lottery**: roughly 30,000 drawn from a list of 85,000.
- Of those selected, about 10,000 enrolled.
- **First stage** (Finkelstein et al. 2012): winning raised the probability of ever being on Medicaid by about **26 percentage points** — large, but nowhere near one. Hence IV rather than a simple comparison.
- **What the design gives you:** the effect of *having Medicaid* for those who enrolled **because** they won — not for those who would have found coverage anyway.

Oregon, in equations

Y_i	utilisation or financial strain — e.g. $\mathbf{1}\{\text{has a usual place of care}\}$
D_i	$\mathbf{1}\{\text{enrolled in Medicaid}\}$
Z_i	$\mathbf{1}\{\text{won the lottery}\}$

$$\begin{aligned}
\text{First stage:} \quad & D_i = \pi_0 + \pi_1 Z_i + \nu_i, \qquad \hat{\pi}_1 \approx 0.26 \\
\text{Reduced form:} \quad & Y_i = \gamma_0 + \gamma_1 Z_i + u_i \\
\text{Second stage:} \quad & Y_i = \alpha + \delta \hat{D}_i + e_i, \qquad \hat{\delta} = \hat{\gamma}_1 / \hat{\pi}_1
\end{aligned}$$

- On the next slide, $\hat{\gamma}_1$ is the **ITT column** and $\hat{\delta}$ is the **LATE column**.

What Oregon found

Outcome	ITT (reduced form)	LATE (2SLS)
Has a usual place of care	9.9%	33.9%
Has a personal doctor	8.1%	28.0%
Got all needed healthcare	6.9%	23.9%
Any hospital admission	0.5%	2.1%
Had a medical collection	−1.6%	−6.4%
Owed on medical collection	−$99	−$390

- The **ITT column is the reduced form** — the policy effect of *running the lottery* — what a legislator decides on.
- The **LATE column scales it by the first stage** — the effect of *coverage* on the outcomes.

2. Judge and Examiner Fixed Effects

The recipe

The design needs three ingredients:

1. A **narrow pipeline** every unit must pass through;
2. **Randomly assigned decision-makers** staffing it;
3. **Discretion** among those decision-makers.

- The canonical setting is American courts, where defendants are randomly assigned to judges.
- **Gaudet, Harris and John (1933)** found one judge imprisoned in 34% of his cases and another in nearly 60%, with cases randomly rotated between them.
- Their conclusion: a sentence may indicate *"either the seriousness of his crime or the severity of the judge."*

One modern example, and how far it travels

- **Stevenson (2018)**, Philadelphia bail magistrates. Randomly assigned judges differ in how affordable they set bail, so some defendants are detained pre-trial **purely by draw**.
- She finds pre-trial detention raises conviction by about **13%**, mostly through guilty pleas, and raises sentence length and court fees.

- The design has spread far beyond courts. Anywhere a queue is staffed by rotating officials with discretion:
  - patent examiners, disability examiners, loan officers, auditors, bankruptcy judges, bank supervisors.

Leniency, in equations

Y_i	convicted $\{0, 1\}$ — also sentence length, court fees
D_i	$\mathbf{1}\{\text{detained pre-trial}\}$
Z_i	leave-one-out detention rate of the judge who drew case i

$$\text{First stage:} \quad D_i = \pi_1 Z_i + \mathbf{c}'_{ct}\gamma_1 + \nu_i$$

$$\text{Second stage:} \quad Y_i = \delta \hat{D}_i + \mathbf{c}'_{ct}\gamma + e_i$$

- $\mathbf{c}_{ct}$ are **court** $\times$ **period fixed effects** — the cell inside which assignment is random. They are not controls; they define the experiment.
- Stevenson's $\hat{\delta}$: detention raises conviction by about 13%.

Building the instrument

Not the judge — the judge's *leniency*: *of all the other cases my judge handled, what fraction did they treat?*

$$Z_i \;=\; \frac{1}{n_{j(i)} - 1} \sum_{\substack{k \neq i \\ j(k) = j(i)}} D_k \qquad \text{(leave-one-out, or } \textit{jackknife}\text{, leniency)}$$

i	the case you are building the instrument *for*
$j(\cdot)$	the judge who drew case i
$n_{j(i)}$	that judge's caseload; subtract 1 because we dropped case i
D_k	whether case k was treated (detained, liquidated, granted)

- A judge who detains 70% of *everyone else* is a judge your case was unlucky to draw — and that luck is the only thing the instrument uses.

Two choices that make it work

- **Why leave your own case out?** Keeping it in puts D_i on *both* sides:

$$Z_i^{\text{keep}} = \underbrace{\frac{1}{n_{j(i)}} \sum_{\substack{k \neq i \\ j(k)=j(i)}} D_k}_{\text{the judge}} + \underbrace{\frac{1}{n_{j(i)}} D_i}_{\text{your own case}}$$

  - Z_i^{keep} then inherits a slice of D_i, and so a slice of the structural error ε_i (how serious the case is). **The instrument is no longer as good as random.**
  - The first stage looks strong for a reason that has nothing to do with the judge.
  - The slice is $1/n_{j(i)}$, so the damage grows when caseloads are short *or* when genuine leniency differences are small.
- **Why residualise first?**
  - Assignment is random *within* a court $\times$ period cell, not nationally.
  - So build leniency inside the group, and carry those fixed effects into both stages.

Where it breaks

- **Independence** — usually fine, assignment really is random.
- **Exclusion** — the usual worry. A defendant who draws a judge known to be severe expects a worse outcome at trial, so pleads guilty — *whether or not he was detained*. That path runs from Z straight to Y, bypassing **detention**, which is the treatment.
- **Monotonicity** — the hardest. Lenient overall, strict on drug charges, and the assumption is gone. Imbens and Angrist flagged this case in 1994: no nonparametric LATE.

Why not just use judge dummies?

- What is random is **which judge you drew**, so 2SLS uses **one dummy per judge** as instruments:

$$D_i \;=\; \sum_{j=1}^{K} \pi_j \, \mathbf{1}\{j(i) = j\} \;+\; \nu_i \;=\; Z_i'\pi + \nu_i$$

- Here Z_i is the vector of K judge dummies. The first stage sets each $\hat{\pi}_j$ to judge j's detention rate over **all** her cases — **yours included**: $\hat{D}_i = Z_i'\hat{\pi} = Z_i^{\text{keep}}$, the instrument we just ruled out.
- Count your case, and the judge's "strictness" moves with **your own outcome**: a serious case gets detained and makes its own judge look strict.
- Leaving it out is the fix. Applied to the dummies' first stage, it is called **JIVE**.

How big is the bias? Look at the extremes

- **One case per judge.** The judge's rate *is* your own outcome: $\hat{D}_i = D_i$. 2SLS **is** OLS, with all of OLS's bias.
- **Thousands of cases per judge.** Your case is a vanishing slice of the rate. 2SLS is the clean IV.
- **In between**, 2SLS sits between the two, pulled **toward OLS** — harder the fewer cases each judge has:

$$\text{2SLS bias} \approx \frac{\text{OLS bias}}{F+1}$$

- F asks whether judges' rates differ more than chance alone would make them. With few cases each, chance alone scatters the rates widely, so F is small. Thin caseloads mean a low F *and* a big slice of your own case — the same fact, twice.

JIVE: leave yourself out

- **The fix:** Estimate the first stage **without case** i, then use it to predict case i. Same K instruments — different fitted value.

$$\hat{D}_i^{\text{JIVE}} \;=\; Z_i' \hat{\pi}_{(-i)}, \qquad \hat{\pi}_{(-i)} \text{ fitted on everyone but } i.$$

- Regress Y_i on $\hat{D}_i^{\text{JIVE}}$. This is **JIVE** (Angrist, Imbens and Krueger 1999).
- Nothing about your own case is used to predict your own case.

- **You do not run n regressions.** Leave-one-out has a closed form, built from the **leverage** of observation i:

$$\hat{D}_i^{\text{JIVE}} \;=\; \frac{Z_i' \hat{\pi} - h_i D_i}{1 - h_i}, \qquad h_i = Z_i'(Z'Z)^{-1} Z_i.$$

- h_i is the ith diagonal of the projection ("hat") matrix — how much case i pulls its own fitted value. See Hansen, *Econometrics*, §3.21.

Your leniency measure *was* JIVE

With judge dummies and nothing else, your fitted value is just your judge's average, $Z_i'\hat{\pi} = \bar{D}_{j(i)}$, and your leverage is your share of that average, $h_i = 1/n_{j(i)}$, where $n_{j(i)}$ is your judge's caseload. So

$$\hat{D}_i^{\text{JIVE}} = \frac{\bar{D}_{j(i)} - D_i/n_{j(i)}}{1 - 1/n_{j(i)}} = \frac{1}{n_{j(i)} - 1} \sum_{\substack{k \neq i \\ j(k) = j(i)}} D_k.$$

- That is **exactly** the leave-one-out leniency we built earlier.
- **Caveat:** Add controls and this breaks — next slide.

Once you add controls, JIVE breaks too

- Assignment is random only *within* a court × period group, so we control for group fixed effects. By FWL, the instrument then becomes **your judge's rate − your group's rate**: how strict your judge is *relative to the other judges in your court*.
- JIVE leaves your case out of the first term only. The group's rate is still computed from every case in the group, yours included.
- So your own case walks back in through the second term. The bias returns.

- **UJIVE** (Kolesár 2013) leaves case *i* out of *both* pieces: your judge's leave-out rate − your group's leave-out rate.
- Without group controls the difference is negligible. With them, UJIVE is the safe default.

Two more things worth knowing

- **Do not cluster by judge.** It is common, and it is wrong. What matters is how cases were *assigned*.
  - Cases assigned to judges one at a time, independently ⇒ plain **robust** standard errors.
  - Cases assigned in batches — one duty judge takes every case filed that day ⇒ cluster **by day**, the unit that was actually drawn.
- **With UJIVE you need not chase a large F.** It stays roughly unbiased even when F is close to 1. That is the point of using it.

- **Read:** Goldsmith-Pinkham, Hull and Kolesár (2026), *Leniency Designs: An Operator's Manual*, *JEP* 40(3). A step-by-step guide — including a weaker, more believable version of monotonicity and a test for it — with an `R` package (`ManyIV`).

Stevenson's numbers (Remix, Table 7.12)

	OLS	OLS	2SLS	2SLS	JIVE	JIVE
Detention	−0.001	0.029	0.151	0.186	0.162	0.212
	(0.002)	(0.002)	(0.065)	(0.064)	(0.070)	(0.076)

$N = 331{,}971$, eight bail judges. Outcome: guilty plea. Odd columns use time controls only; even columns add defendant and offence characteristics.

- **OLS says detention does nothing.** IV says it raises guilty pleas by 15–21 percentage points.
- But Stevenson has only **eight** judges. To see what happens with *many*, we need a bigger pipeline.

A bigger pipeline: patent examiners

Goldsmith-Pinkham, Hull and Kolesár (2026) re-run Farre-Mensa, Hegde and Ljungqvist (2020): does getting your **first patent approved** help a startup innovate again?

- **Treatment** D_i: the application was **approved**.
- **Instrument** Z_i: **one dummy per examiner** — examiners differ in how freely they grant.
- **Outcome** Y_i: the startup's later innovation
- **Controls** W_i: **art unit** $\times$ year.
  - An *art unit* is the small group of examiners who specialise in one technology; an application is assigned quasi-randomly to an examiner **within** it. It is the patent office's court-and-period group — the place where the lottery actually happens.

The standard errors lie too (GPHK 2026)

Effect of approval on	UJIVE	2SLS (examiner dummies)	OLS
Any later application	0.173	0.232	0.234
	(0.055)	(0.016)	(0.006)
Any later approval	0.259	0.240	0.223
	(0.050)	(0.014)	(0.005)
$\log(1 + \text{citations})$	0.419	0.372	0.339
	(0.125)	(0.033)	(0.011)

- **2SLS has landed on OLS** — That is the bias reproduced.
- **And look at the standard errors.** 2SLS reports errors three to four times *smaller* than UJIVE.

3. Bartik / Shift-Share IVs

The question: what does immigration do to local wages?

- **The question.** When immigrants arrive in a city, what happens to the wages of the **native** workers already there?

$$Y_{l,t} = \alpha + \delta D_{l,t} + \rho X_{l,t} + \varepsilon_{l,t}$$

- $Y_{l,t}$: log native wages in city l in year t. $D_{l,t}$: immigrant inflow. $X_{l,t}$: controls, including city and year fixed effects.
- **Why OLS cannot answer it.** Immigrants are not sprinkled over cities at random. They go where the work is.
- **What we need:** something that sends immigrants to a city for reasons that have nothing to do with *that city's* labour market.

The idea: you go where your people already are

- A new arrival from Mexico does not choose a city off a map. They go where there is already a Mexican community.
- So a **national** surge in migration from one country lands disproportionately in the cities that already held people from that country.
- l = a **city**, k = a **source country**, t = the year, t_0 = a base year *before* the flows we study.

$$\underbrace{B_{l,t}}_{\text{predicted inflow}} = \sum_{k=1}^{K} \underbrace{z_{l,k,t_0}}_{\substack{\textbf{shares} \\ \text{who was already there}}} \times \underbrace{m_{k,t}}_{\substack{\textbf{shifts} \\ \text{who left home}}}$$

- Los Angeles chose neither the surge out of Mexico nor, in anticipation of it, its 1980 immigrants. The surge was decided in Mexico.

Shift-share, in equations

$Y_{l,t}$	log wages of *native* workers in city l
$D_{l,t}$	immigrant inflow into city l — the Remix writes this $I_{l,t}$
$Z_{l,t}$	$B_{l,t} = \sum_k z_{l,k,t_0}\, m_{k,t}$ — inflow predicted by the enclave map
$X_{l,t}$	controls: city and year fixed effects

$$\text{First stage:} \quad D_{l,t} = \pi_0 + \pi_1 B_{l,t} + \rho_1 X_{l,t} + \nu_{l,t}$$

$$\text{Second stage:} \quad Y_{l,t} = \alpha + \delta \hat{D}_{l,t} + \rho X_{l,t} + e_{l,t}$$

- First stage: do the cities whose enclaves *should* have drawn immigrants actually get them?
- δ is the effect of immigration on native wages — an average across source countries, weighted towards whichever ones actually move the instrument.

Identification

Bartik IV requires exogeneity of either shares OR shifts.

Framework A: Shares Exogenous

Goldsmith-Pinkham, Sorkin, Swift (GPSS, 2020)

- Identifies via baseline shares z_{ik}.
- Equivalent to using shares as individual instruments.
- Shifts $m_{k,t}$ act merely as a weighting matrix.

Framework B: Shifts Exogenous

Borusyak, Hull, Jaravel (BHJ, 2022)

- Identifies via sectoral shocks m_{kt}.
- Shares z_{ik} can be completely endogenous.
- Requires many independent shocks ($K \to \infty$).

How identification fails: Framework A (exogenous shares)

$Y_{l,t}$ = wage growth in city l, $D_{l,t}$ = local employment growth (labour demand), and

$$Z_{l,t} = \sum_{k=1}^{K} \underbrace{z_{l,k,1990}}_{\substack{\text{city } l\text{'s 1990 employment} \\ \text{share in industry } k}} \times \underbrace{m_{k,t}}_{\substack{\text{national employment} \\ \text{growth of industry } k}}$$

- **Path dependence.** Cleveland's 1990 manufacturing share is high, so its Z moves whenever national manufacturing moves. But Rust Belt cities were already on a declining wage path for reasons that have nothing to do with this year's national shock — the share is correlated with $\varepsilon_{l,t}$.
- **Agglomeration and amenities.** Seattle's tech share is high, so its Z loads on national tech growth. But a tech-heavy city also has educated workers, costly housing and venture capital, which move wages directly. The share is a proxy for all of that.

How identification fails: Framework B (exogenous shocks)

Same $D_{l,t}$ and the same $Z_{l,t} = \sum_k z_{l,k,1990}\, m_{k,t}$.

- **Common macro shocks.** $Y_{l,t}$ = growth in local consumer spending. National auto employment jumps because the Fed cut rates. The shift $m_{\text{auto},t}$ is then not an auto-industry shock; it *is* the rate cut. Auto-heavy cities' spending responds to the rate cut directly, so $m_{k,t}$ is correlated with $\varepsilon_{l,t}$ exactly in the cities where $Z_{l,t}$ is large.
- **Few dominant shocks.** $Y_{l,t}$ = local wage growth, $K = 50$ industries — but oil & gas and healthcare account for most of the variation in $m_{k,t}$. You effectively have two instruments, not fifty. BHJ needs a law of large numbers across k so that any one shock's endogeneity washes out; with two, whatever confounds oil or healthcare *is* the estimate.

Choose your framework first

- **State your framework:** exogenous shares (GPSS) or exogenous shocks (BHJ).
- The framework determines the diagnostics:
  - GPSS: Rotemberg weights; balance and pre-trends for the top-weighted industries.
  - BHJ: shock-level balance; effective number of shocks; exposure-robust SEs.

- **Read:** Breuer (2022), *Bartik Instruments: An Applied Introduction*, *JFR* 7(1). Written for accounting and finance.
- **Read:** Borusyak, Hull and Jaravel (2025), *A Practical Guide to Shift-Share Instruments*, *JEP* 39(1).

4. Fuzzy Regression Discontinuity

Lecture 4, revisited

- In a **sharp** design, $D_i = \mathbf{1}\{X_i \geq c_0\}$ exactly. In a **fuzzy** design, crossing the cutoff raises the probability of treatment without forcing it:

$$\lim_{x \downarrow c_0} \Pr(D_i = 1 \mid X_i = x) \neq \lim_{x \uparrow c_0} \Pr(D_i = 1 \mid X_i = x).$$

- The estimator is a **Wald ratio** of two discontinuities:

$$\delta_{\text{FRD}} = \frac{\lim_{x \downarrow c_0} E[Y|X = x] - \lim_{x \uparrow c_0} E[Y|X = x]}{\lim_{x \downarrow c_0} E[D|X = x] - \lim_{x \uparrow c_0} E[D|X = x]}.$$

- A ratio of two discontinuities is a Wald estimator. So a fuzzy RD **is** an IV design.

Fuzzy RD is an IV design

- Fuzzy RD **is** IV, with the instrument $Z_i = \mathbf{1}\{X_i \geq c_0\}$:
  - Numerator = reduced-form jump in Y; denominator = first-stage jump in D.
  - Estimate by 2SLS within a bandwidth, instrumenting D with Z and controlling for a local polynomial in X on each side.
  - Same IV assumptions, holding locally at c_0: crossing the cutoff affects Y only through D (exclusion), and no one is pushed *out* of treatment by crossing it (monotonicity). Relevance. SUTVA.
  - So δ_{FRD} is a LATE for **compliers at the cutoff**: units whose treatment is switched by crossing c_0.

Fuzzy RD, in equations

Y_i	the outcome
D_i	treatment actually **received** — not assignment
Z_i	$\mathbf{1}\{X_i \geq c_0\}$ — crossing the cutoff

$$\text{First stage:} \quad D_i = \pi_0 + \pi_1 Z_i + f(X_i - c_0) + \nu_i$$

$$\text{Second stage:} \quad Y_i = \alpha + \delta \hat{D}_i + g(X_i - c_0) + e_i$$

- f and g are the running-variable polynomials, fitted **separately on each side** and only inside the bandwidth, with Lecture 4's kernel weights.
- δ is a LATE *at the cutoff*, for units whose treatment flips there — doubly local.

5. The Elasticity of Demand

The Identification Problem

- **The Target:** A demand curve is a causal object—the effect of price (P) on quantity demanded (Q).
- **The Obstacle:** Observed (P, Q) pairs are **equilibrium points** generated by simultaneous shifts in both supply and demand.
- Wright's Solution:
  - **History:** Instrumental variables were invented for this exact problem by **Philip Wright (1928)**.
  - **The Logic:** To trace out a fixed demand curve, you need exogenous variation that shifts **supply only**.

The question: how price-sensitive are fish buyers?

- **The question.** What is the **price elasticity of demand** for fish? **Kathryn Graddy** set out to measure it at the **Fulton Fish Market**. **She hand-collected the data**, driving in from New Jersey before dawn each morning.
- To hold the good fixed, she narrowed it to one product: **fresh whiting of roughly the same size.**

- **The obstacle is Wright's.** Each trading day hands you a single (P, Q) pair, set by supply and demand *at once*. Regressing $\log Q$ on $\log P$ traces neither curve.
- **The instrument:** the weather at sea in the days *before* the fish reached the market. (Graddy 1995, 2006)

Fulton Fish, in equations

Y_t	$\log Q_t$ — log pounds of whiting sold on day t
D_t	$\log P_t$ — log average price that day
Z_t	$\texttt{Stormy}_t$ — bad weather at sea over the previous two days

$$\text{First stage:} \quad \log P_t = \pi_0 + \pi_1 \texttt{Stormy}_t + \mathbf{d}_t'\gamma_1 + \nu_t, \qquad \hat{\pi}_1 \approx 0.35$$

$$\text{Second stage:} \quad \log Q_t = \alpha + \delta \widehat{\log P_t} + \mathbf{d}_t'\gamma + e_t, \qquad \hat{\delta} \approx -1.12$$

- $\mathbf{d}_t$ are day-of-week dummies. The unit is a **day**, not a buyer.
- Both variables are in **logs**, so δ *is* the elasticity — no conversion required.

Fulton Fish Market (Remix, Table 7.14)

Dependent variable: log Q	OLS	2SLS
log(Price)	-0.563^{***}	-1.119^{**}
	(0.152)	(0.443)
		$[-2.273,\ -0.312]$
First stage		
`Stormy`		0.346^{***}
		(0.072)
Effective F		22.929
N	111	111

Graddy (2006), 111 days at the Fulton Fish Market. Q = log pounds of whiting sold, P = log average daily price. `Stormy` records bad weather at sea over the previous two days. Robust standard errors in parentheses, Anderson–Rubin 95% interval in brackets. All models control for day-of-week dummies. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Reading the fish

- Stormy weather raises price by about **35%**. That is a **supply** shock, not a demand shock — storms make fish harder to catch, they do not make New Yorkers hungrier.
- **OLS says the elasticity is** -0.56**. IV says** -1.12 — twice as elastic.

- **Why does OLS understate it?** Because the observed price–quantity pairs are *equilibria*. Both curves move, so the scatter traces out neither one.

Even here, it is only a LATE

- The IV elasticity of -1.12 says: **for compliers**, a 10% storm-induced price rise cuts quantity by about 11%.
- Now suppose a seller wants to use this number to set a profit-maximising price. Demand looks elastic, so cut prices and sell more — right?
- **No.** That elasticity was identified by *storms*. When the firm cuts its price deliberately, it is not using a storm.
- Buyers read a storm-driven price rise as **temporary** and a posted price cut as **permanent**. Different shock, different response, different elasticity.

What a LATE is, and is not

- This is the sobering version of LATE:

It does not tell you what you want to know.
It tells you what you can learn, given your instrument.

- The warning applies to **every** design — the lottery, the judge, the national shock, the threshold, the weather.
- Before reporting a LATE, ask one question: *would the policy I have in mind move the same people, in the same way, as my instrument did?*
- If the answer is no, you have a clean estimate of something other than what you were asked.

What the five have in common

Design	The shock	Typical first stage	Hardest assumption
Lotteries	randomised offer	strong	exclusion (usually fine)
Judge FE	assigned decision-maker	many weak instruments	**monotonicity**
Bartik	national shock $\times$ local share	strong	exclusion (shares or shifts)
Fuzzy RD	crossing the cutoff	jump in $\Pr(D = 1)$	exclusion at the cutoff
Elasticity	supply shifter	price response	exclusion (demand channel)

1. **Instruments are physical events.** A lottery draw, a shift schedule, a national policy, a storm. If you cannot describe the physical event that assigns treatment in the real world, you do not have an IV.
2. **Estimands are Local (LATE).** An IV only identifies the treatment effect for the **compliers**—the specific people moved by that exact shock.

- Always name the compliers in plain English before interpreting your estimate.

IV in Finance

Where the shocks are

Finance has fewer clean lotteries than labour economics. The recurring sources are:

- **Regulatory and legal changes** that bind some firms and not others — staggered state laws, index or listing rules, supervisory thresholds.
- **Weather, geography and distance** — historically popular, increasingly contested, because geography correlates with almost everything.
- **Shift-share exposure** — a firm's or region's pre-existing portfolio of industries, suppliers or customers, interacted with aggregate shocks.
- **Randomly assigned decision-makers** — bankruptcy judges, bank examiners, auditors, analysts, patent examiners.
- **Lotteries and quasi-lotteries** — oversubscribed allocations, IPO allocation rules, random audits.

- In finance the agents are **sophisticated, well advised and highly motivated**, so the shock you think is exogenous is often anticipated, lobbied for, or selected into.

Three questions to ask of any finance IV

1. **Is the shock really a shock to someone?** Name the units that were moved and the units that were not, and say why the difference was not their choice. If you cannot describe the assignment mechanism in a sentence a practitioner would recognise, you do not have one.
2. **Is there exactly one channel?** This is where most finance IVs fail. A legal change that alters takeover exposure also alters managerial incentives, disclosure, and the composition of shareholders. Distance to a city affects financing *and* labour markets, customers and information.
3. **Who are the compliers, and do you care about them?** A regulation that binds only small, constrained firms identifies an effect for small, constrained firms — which may be the opposite of the effect for the firms the policy debate is about.

The most useful habit

- Write the exclusion restriction down as a **sentence, in words** — not as an equation, and not as a citation.
- Show it to someone who knows the **institution** and not the econometrics.

- If they can name a second channel in under a minute, you have your answer.
- If you cannot write the sentence at all, you did not have an instrument — you had a variable that happened to be correlated with your treatment.

- Read: Jiang, W. (2017). Have Instrumental Variables Brought Us Closer to the Truth? *Review of Corporate Finance Studies* 6, 127–140.

References

- Angrist, J., K. Graddy and G. Imbens (2000). The Interpretation of IV Estimators in Simultaneous Equations Models. *ReStud* 67, 499–527.
- Angrist, J., G. Imbens and A. Krueger (1999). Jackknife Instrumental Variables Estimation. *Journal of Applied Econometrics* 14, 57–67.
- Baicker, K. et al. (2013). The Oregon Experiment — Effects of Medicaid on Clinical Outcomes. *NEJM* 368, 1713–1722.
- Bartik, T. (1991). *Who Benefits from State and Local Economic Development Policies?* Upjohn Institute.
- Borusyak, K., P. Hull and X. Jaravel (2022). Quasi-Experimental Shift-Share Research Designs. *ReStud* 89, 181–213.
- Borusyak, K., P. Hull and X. Jaravel (2025). A Practical Guide to Shift-Share Instruments. *JEP* 39(1), 181–204.
- Breuer, M. (2022). Bartik Instruments: An Applied Introduction. *Journal of Financial Reporting* 7(1), 49–67. — **read this before building a shift-share instrument.**
- Cunningham, S. *Causal Inference: The Remix*, §7.7 — this deck follows it closely.
- Farre-Mensa, J., D. Hegde and A. Ljungqvist (2020). What Is a Patent Worth? Evidence from the U.S. Patent "Lottery". *Journal of Finance* 75(2), 639–682.
- Finkelstein, A. et al. (2012). The Oregon Health Insurance Experiment: Evidence from the First Year. *QJE* 127, 1057–1106.
- Gaudet, F., G. Harris and C. St. John (1933). Individual Differences in the Sentencing Tendencies of Judges. *Journal of Criminal Law and Criminology* 23, 811–818.
- Goldsmith-Pinkham, P., P. Hull and M. Kolesár (2026). Leniency Designs: An Operator's Manual. *JEP* 40(3), 213–240. — **read this before running a leniency design.**
- Goldsmith-Pinkham, P., I. Sorkin and H. Swift (2020). Bartik Instruments: What, When, Why, and How. *AER* 110, 2586–2624.
- Graddy, K. (1995). Testing for Imperfect Competition at the Fulton Fish Market. *RAND Journal of Economics* 26(1), 75–92.
- Graddy, K. (2006). Markets: The Fulton Fish Market. *JEP* 20, 207–220.
- Hansen, B. (2022). *Econometrics*. Princeton University Press. (§3.21, leave-one-out and leverage.)
- Imbens, G. and J. Angrist (1994). Identification and Estimation of Local Average Treatment Effects. *Econometrica* 62, 467–475.
- Kolesár, M. (2013). Estimation in an Instrumental Variables Model with Treatment Effect Heterogeneity. Working paper. (UJIVE.)
- Jiang, W. (2017). Have Instrumental Variables Brought Us Closer to the Truth? *Review of Corporate Finance Studies* 6, 127–140.
- Stevenson, M. (2018). Distortion of Justice: How the Inability to Pay Bail Affects Case Outcomes. *Journal of Law, Economics, and Organization* 34, 511–542.